



$n = 4, \text{row} = 1, \text{star} = 1,$
 $\text{space} = n - 1$

Total no. of rows = 4

While (row $\leq$ n)

Space -- ;
 star = star + 2 ;
 row ++ ;

While (row $\leq$ n) {

// space

int i = 1 ;

while (i $\leq$ space) {

cout << " " ;

i ++ ;

}

// star

int j = 1 ;

while (j $\leq$ star) {
 cout << " * " ;
 j ++ ;
 }

n=5

```

      n=3 | - *
            * * *
            * * * *
            * * * * *
            * * *
            *
  
```

$\leftarrow \text{row} = 1$
 $\leftarrow \text{row} = 2$
 $\leftarrow \text{row} = 3 \leftarrow$
 $\leftarrow \text{row} = 4$
 $\leftarrow \text{row} = 5$

$\text{row} = 1$ while $(\text{row} \leq 2 * n - 1) \{$
 if $(\text{row} < n)$
 $\leftarrow (\text{row} = n)$

Total rows = 5
 $\text{row} = 1$ & (3)
 Star = * * * 3
 Space = 2 * 0 1

```

// Space
int i = 1;
while (i <= space) {
    cout << " ";
    i++;
}
int j = 1;
while (j <= star) {
    cout << " * ";
    j++;
}
  
```

```

// mirror
if (row < n) {
    star -= 2;
    space--;
} else {
    star += 2;
    space++;
}
cout << endl;
row++;
}
  
```


1 → row 1
 2 2 → row 2
 3 3 3 → row 3
 2 2 2 → row 4
 1 → row 5

Given:- $n=5$, total rows = 5
 row = 1, mirror = 3

Print:-
 Outer loop ($row \leq n$)
 Changing parameter →

↑ row → $star = star + 2$
 ↓ row → $star = star - 2$
 $space = space - 1$
 $space = space + 1$

```

int star = 1, row = 1, space = n/2, n = 5, val = 1;
while (row <= n) {
    // space
    // star
    int j = 1;
    while (j <= star) {
        cout << val << " ";
        j++;
    }
    if (row <= n/2 + 1) {
        val++;
    } else {
        val--;
    }
    row++;
}
  
```




```
while (row <= n) {
```

```
// space
```

```
// star
```

```
int j = 1;
```

```
int num = val;
```

```
while (j <= star) {
```

```
    print(num)
```

```
    num++
```

```
    j++;
```

```
// mirror
```

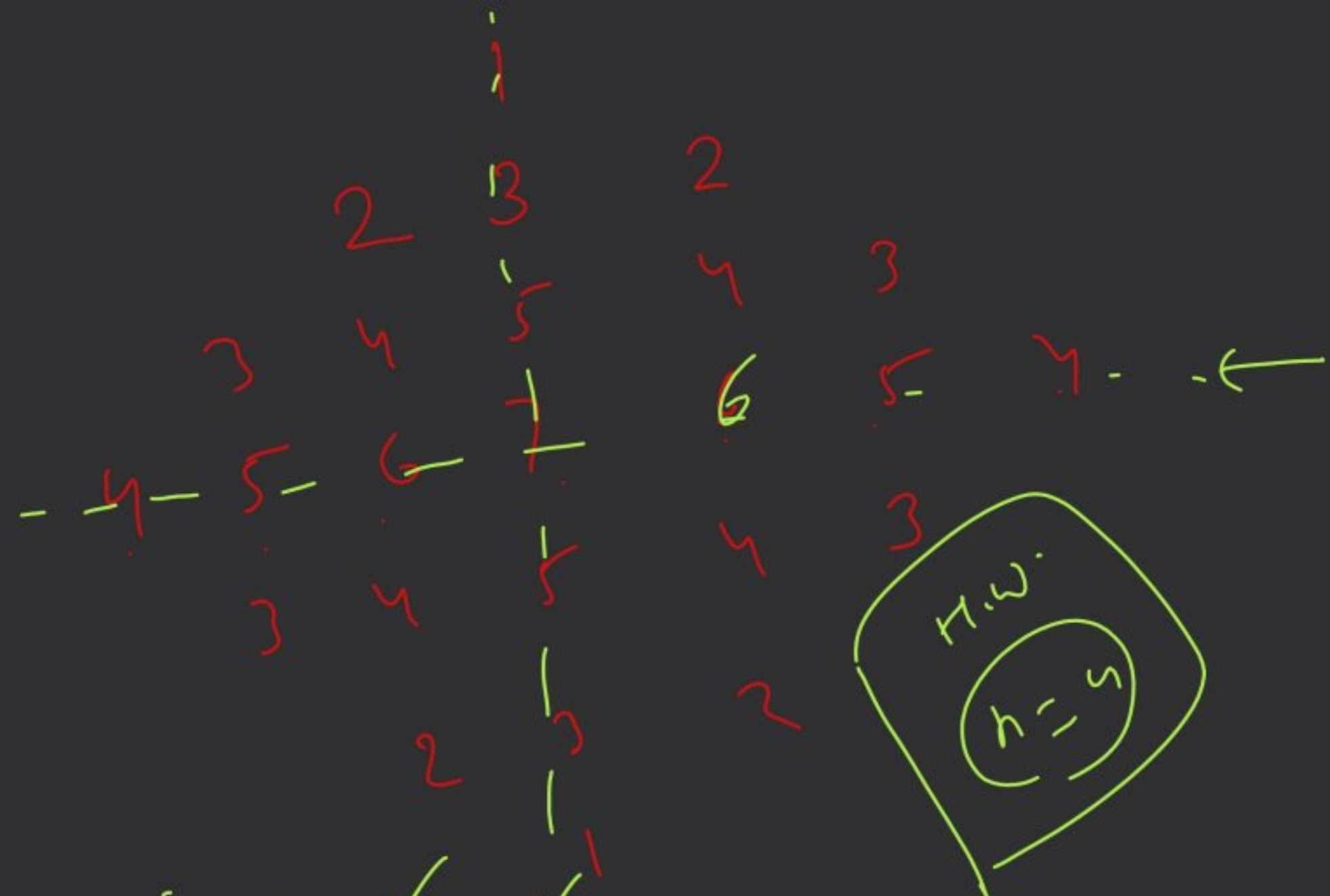
```
    printn();
```

```
    row++;
```

```
}
```



row →	1	2	3	4
star →	2	2	2	2
space →	0	1	2	3
val →	1	2	3	4



$j = x$ ~~2~~ ~~3~~
 $num = 4$ ~~5~~ ~~6~~ ~~7~~

$val = 4$
 $num = 4$

while ($j \leq star$) {

Sort (num);

if ($j < \frac{star}{2} + 1$) {

num++;

else num--;

}
j++;

star = 7

$j < 4$
 $j < 3$
 $\frac{7}{2} = 3$ $j < 3+1$
 $j < \frac{star}{2} + 1$